

Mohamed Abdelhady Ahmed, Ph.D.

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Professional Summary

Summary: Robotics and AI researcher with a Ph.D. in Electrical Engineering (Dec 2021) and research experience spanning representation learning, system identification, state estimation, reinforcement learning, neural networks, wearable sensing, and model-based control of dynamical robotic systems. Developed learning-based methods for prostheses, pediatric exoskeletons, and human movement estimation using multimodal sensor data, including attention-based deep RNNs for EMG-driven state estimation and deep reinforcement-learning controllers for pediatric exoskeletons. Experienced in designing experimental systems, data-driven models, and hardware-in-the-loop demonstrations. Published in IEEE and robotics/biomedical venues and inventor on multiple U.S. patent applications. Professional Python practitioner (PyTorch, TensorFlow, Scikit-Learn). Former Cleveland Clinic Biomedical Engineering postdoctoral fellow (2021-2022); current Adjunct Assistant Professor at Cleveland State University.

Education

Ph.D. in Electrical Engineering 2015 – 2021

Cleveland State University • Cleveland, OH

Advisor: Prof. Dan Simon

Thesis: “Codesign Development and Control of Smart Prostheses”

Dissertation: Codesign Development and Control of Smart Prostheses.

M.Sc. in Computers & Systems Engineering 2010

Minia University • Minia, Egypt

Thesis: “Deep Insight in Control of Twin Rotor MIMO Helicopter System”

B.Sc. in Electrical Engineering (Control Engineering) 2001

Aswan University • Aswan, Egypt

Graduated Summa Cum Laude.

Academic Positions

Adjunct Assistant Professor 2024 – Present

Cleveland State University • Electrical Engineering & Computer Science • Cleveland, OH

Teach courses including Artificial Intelligence for Engineers, Biomedical Signal Processing, Robotics and Intelligent Systems, and Engineering Programming (MATLAB & Python).

Postdoctoral Fellow & Associate Investigator 2022 – 2024

National Institutes of Health (NIH), Clinical Center • Rehabilitation Medicine • Bethesda, MD

Deep learning and reinforcement learning for personalized pediatric exoskeleton control and gait assessment.

Postdoctoral Fellow 2021 – 2022
Cleveland Clinic (Lerner Research Institute) • Biomedical Engineering • Cleveland, OH

Data-driven detrusor pressure estimation, physiological signal processing, and clinical NLP pipelines; HIPAA-compliant.

Research Collaborator 2021 – 2022
U.S. Department of Veterans Affairs, Advanced Platform Technology Center • Cleveland, OH

Estimation and signal-processing methods for single-channel physiological sensing and pressure reconstruction.

Research Engineer 2019 – 2020
Ford Motor Company • Detroit, MI

Neural-network control for vehicle performance and fuel efficiency; real-time control systems.

Research Experience

Position/role: Postdoctoral Fellow & Associate Investigator
Institution: National Institutes of Health (NIH), Clinical Center
Advisor: Dr. Thomas C. Bulea
Start year: 2022
End year: 2024

Description: Developed deep learning and reinforcement learning methods for personalized pediatric exoskeleton control, modeling time-varying human movement and system behavior. Developed neural-network methods for gait and lower-limb state estimation from wearable physiological sensors (EMG-driven knee angle estimation, attention-based deep RNNs). Designed and evaluated a pediatric exoskeleton integrating robotic assistance with functional electrical stimulation. Designed controlled experiments evaluating learned models for accuracy, robustness, and downstream robotic control. Explored AI-assisted clinical workflows using NLP and LLMs for healthcare knowledge extraction.

Position/role: Postdoctoral Fellow, Biomedical Engineering
Institution: Cleveland Clinic (Lerner Research Institute)
Advisor: Drs. Margot Damaser & Steve Majerus
Start year: 2021
End year: 2022

Description: Developed data-driven state/parameter estimation algorithms for detrusor pressure from single-channel bladder pressure recordings; clinically validated using real patient data. Applied physiological signal processing and machine learning to improve estimation from limited sensor observations. Designed NLP pipelines extracting structured information from clinical documentation and EHR data; collaborated with clinicians on decision-support tooling; HIPAA-compliant. Served as Research Collaboration Coordinator across engineering, clinicians, and students.

Position/role: Research Collaborator
Institution: U.S. Department of Veterans Affairs, Advanced Platform Technology Center
Advisor: Drs. Margot Damaser & Steve Majerus
Start year: 2021

End year: 2022

Description: Estimation and signal-processing methods for single-channel physiological sensing and pressure reconstruction; co-inventor on resulting U.S. patent applications.

Position/role: Ph.D. Researcher, Electrical Engineering

Institution: Cleveland State University

Advisor: Prof. Dan Simon

Start year: 2015

End year: 2021

Description: Dissertation: Codesign Development and Control of Smart Prostheses.

Developed learning-based models of lower-limb prostheses (system identification, neural networks, reinforcement learning) and reinforcement-learning impedance control.

Developed a wearable multi-sensor system for lower-limb kinetics/kinematics and reinforcement-learning impedance control for prostheses.

Designed hardware-in-the-loop experiments; fabricated prosthesis prototypes via additive manufacturing.

Developed wireless IoT wearable sensing and embedded data-acquisition systems for continuous gait monitoring.

Position/role: Research Engineer

Institution: Ford Motor Company

Start year: 2019

End year: 2020

Description: Neural-network control and real-time signal-processing systems; presented internal ML research at Ford's AI/ML conference.

Position/role: Engineering Intern

Project: 3D reconstruction for MRI; human-assistive robotics

Institution: Siemens

Advisor: Dr. Ivan Fischer

Start year: 2011

Description: 3D reconstruction for MRI; human-assistive robotics with Festo.

Publications

M. Abdelhady & T. C. Bulea (2025). *Gaussian regressor-based adaptive control of exoskeleton joints in the presence of system uncertainty*. *Wearable Technologies*, 6, e44.

S. Zhang, M. Lopez-Sanchez, **M. Abdelhady** (2024). *Actuator optimization and deep learning-based control of pediatric knee exoskeleton for community-based mobility assistance*. *Mechatronics*, 97, 103109.

M. Abdelhady, D. L. Damiano, T. C. Bulea (2024). *Knee angle estimation from surface EMG during walking using attention-based deep recurrent neural networks*. *Sensors*, 24(13), 4217.

M. Abdelhady, A. J. van den Bogert, D. Simon (2019). *A high-fidelity wearable system for measuring lower-limb kinetics and kinematics*. *IEEE Sensors Journal*, 19(24), 12482-12493.

E. Elsakka, **M. Abdelhady**, S. Anwar (2022). *Depression and anxiety among siblings of children with cerebral palsy*. *Journal of the High Institute of Public Health*, 52(1), 24-32.

- M. Subramanian & **M. Abdelhady** (2022). *The role of contemporary digital tools and technologies in Covid-19 crisis*. Expert Systems, 39(6), e12834.
- V. Sathishkumar & **M. Abdelhady** (2021). *Secure dynamic group data sharing in semi-trusted third-party cloud environment*. Arabian Journal for Science and Engineering, 1-9.
- M. Abdelhady**, M. El-Sayed, M. Mohamed (2009). *PID controller tuning for twin rotor MIMO system based on particle swarm optimization*. Journal of Engineering Sciences, 37(4), 955-967.
- M. Abdelhady** & D. Simon (2025). *Application of superimposed neural network in human gait estimation*. IEEE Transactions on Systems, Man, and Cybernetics: Systems. [Accepted]
- M. Abdelhady**, T. C. Bulea, D. Simon (2026). *Synergizing assistive robotics and muscle electrical stimulation*. IEEE Transactions on Biomedical Engineering. [Under Review]
- M. Abdelhady**, S. Majerus, M. S. Damaser (2026). *Advanced techniques for detrusor pressure estimation*. IEEE Transactions on Biomedical Engineering. [Under Review]
- M. Abdelhady** & T. C. Bulea (2023). *Attention-based deep RNN to estimate knee angle during walking from lower-limb EMG*. IEEE International Conference on Rehabilitation Robotics (ICRR).
- M. Abdelhady** & T. C. Bulea (2023). *Prediction accuracy and model robustness of neural network-based ground reaction force estimators*. IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM).
- M. Abdelhady**, S. Majerus, M. S. Damaser (2022). *Event-driven detrusor pressure estimator for single catheter applications*. IEEE Signal Processing in Medicine and Biology Symposium (SPMB), 503-509.
- M. Abdelhady** & D. Simon (2020). *Prosthesis controller hardware-in-the-loop simulation*. 3rd International Conference on Control and Robots.
- M. Abdelhady**, Y. P. Kondratenko, D. Simon (2020). *Reinforcement learning for impedance control for lower-limb prostheses*. Automation 2020, 115-116.
- M. Abdelhady** & D. Simon (2017). *System identification and control optimization of an active prosthetic knee in swing phase*. American Control Conference (ACC), 857-862.
- M. Abdelhady**, Y. P. Kondratenko, D. Simon (2019). *Stability analysis of heartbeat control based on the Zeeman framework*. IEEE International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications (IDAACS), 824-829.
- M. Abdelhady**, A. Elsheikh, M. Ahmed (2021). *Design a compact wireless IoT gait monitor wearable sensory system*. International Telecommunication Conference (ITC), 1-4.
- Margot S. Damaser, Steve Majerus, **Mohamed Abdelhady** (2022). *Detrusor pressure estimation from single channel bladder pressure recordings*. U.S. Patent and Trademark Office - U.S. Patent Application No. 17/880,109.

Teaching Experience

Course: Multiple courses (20+)

Institution: Cleveland State University

Role: Instructor

Description: Taught more than 20 courses including Artificial Intelligence for Engineers, Biomedical Signal Processing, Robotics and Intelligent Systems, and Engineering Programming (MATLAB & Python).

Awards and Honors

Award: Cleveland Clinic Travel Grant (AUA Conference)

Organization: Cleveland Clinic

Year: 2022

Award: Cleveland Clinic Fellowship - ML & signal processing for bladder biometrics

Organization: Cleveland Clinic

Year: 2022

Award: Best Paper Award - Event-driven detrusor pressure estimator

Organization: IEEE Signal Processing in Medicine and Biology Symposium (SPMB)

Year: 2022

Award: Student Research Award (outstanding Ph.D. candidate)

Organization: Cleveland State University

Year: 2021

Award: Best Paper Award - Prosthesis controller hardware-in-the-loop simulation

Organization: 3rd International Conference on Control and Robots

Year: 2020

Award: Excellent Oral Presentation Award

Organization: ICCR Conference

Year: 2020

Award: Dean Travel Award

Organization: Cleveland State University

Year: 2017

Award: Graduate School Award

Organization: Cleveland State University

Year: 2017

Award: Egyptian Government Scholarship

Organization: Egyptian Government

Year: 2014

Grants and Fellowships

Grant/fellowship name: Egyptian Government Scholarship

Organization: Egyptian Government

Role: Other

Start year: 2014

Description: Fully funded scholarship for Ph.D. studies.

Grant/fellowship name: Cleveland Clinic Fellowship - bladder biometrics ML project

Organization: Cleveland Clinic

Role: Postdoctoral Fellow

Start year: 2022

End year: 2022

Description: Machine learning and signal processing for bladder biometrics.

Academic Service

Organization: Expert Systems with Applications

Role: Peer Reviewer

Organization: Engineering Applications of Artificial Intelligence

Role: Peer Reviewer

Organization: Applied Soft Computing

Role: Peer Reviewer

Organization: IEEE Control Letters

Role: Peer Reviewer

Organization: IEEE Robotics & Automation Society

Role: Member

Ongoing: Yes

Organization: IEEE Control Systems Society

Role: Member

Ongoing: Yes

Organization: International Society of Automation

Role: Member

Ongoing: Yes

Organization: Cleveland State University

Role: ABET Accreditation & Assessment Coordinator; ECE Curriculum Committee Member

Organization: Cleveland State University

Role: Founder & Coordinator, Intelligent Systems, Robotics & AI Research Group

Organization: Cleveland State University

Role: Advisor, IEEE / Robotics / AI student chapters

Skills

Programming: Python (PyTorch, TensorFlow, Keras, Scikit-Learn, NumPy, pandas, SciPy, XG-Boost), MATLAB, C++, C#, Java, SQL, Git

Deep Learning: RNN/LSTM, CNN, attention models, representation learning, transfer learning, model robustness & interpretability analysis

LLMs & NLP: Large language models, RAG, agentic AI, prompt engineering, clinical/EHR information-extraction pipelines, model evaluation

Signal Processing: ECG, EMG, EEG, digital filtering, wavelet & time-frequency analysis, feature extraction, time-series analysis

Medical Imaging: Biomedical image classification, image preprocessing, quantitative image analysis, CNN-based recognition

Reinforcement Learning: RL for control, impedance control, policy learning, simulation (MuJoCo, Isaac Lab, Gazebo)

Robotics & Control: Model predictive control, adaptive/nonlinear control, optimal control, hardware-in-the-loop

Wearable Sensing: EMG, IMU, gait monitoring, exoskeletons, prostheses, embedded data acquisition

Systems & Tools: Linux, Windows, ROS/ROS2, Docker, MLflow, Jupyter, AWS/Azure, EHR data pipelines, HIPAA-compliant data governance

Engineering: Sensor fusion, wearable sensor networks, hardware-in-the-loop testing, additive manufacturing, SolidWorks, Altium